ETHNOGRAPHIC STUDY IN
INTERACTION DESIGN

ECHOES OF PERCEPTION

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OBJECTIVE OF THE SURVEY



The objective of this survey was to examine how a single piece of music influences people's emotions, imagination, spatial perception, and associative thinking. The study aimed to understand participants' emotional state before listening and observe whether the song triggered any emotional shift. It also explored the types of imagery, memories, or situations the music evoked, and how individuals connected the song to elements such as brands, food, workplaces, and environments. Additionally, the research analyzed how participants perceived the song in terms of intensity, pace, temperature (hot or cold), and spatial scale. Overall, the purpose of the study was to understand how music shapes psychological perception and sensory interpretation across different personalities and backgrounds.

METHODS USED

The study used a primary research method through a structured questionnaire.

It was a quantitative + qualitative mixed method approach:

- **Quantitative** → Multiple choice questions for measurable data (mood, pace, intensity, etc.)
- **Qualitative** → Association-based questions (imagery, brands, food, workplace)

Tools Used - The data for this study was collected using a structured questionnaire created through Google Forms. Participants were asked to listen to the selected song using headphones to ensure clear audio quality and minimize external distractions. They were instructed to listen to the complete duration of the song before responding to the questions. The responses were automatically recorded through Google Forms and later organized using spreadsheet software such as Google Sheets for data tabulation. Percentage analysis and pie charts were then used to visually represent and interpret the results.

PROCEDURE OF ANALYSIS

- Participants were first asked about their mood before listening.
- They were then asked to listen to the song completely without distraction.
- Immediately after listening, they answered the questionnaire.
- Responses were recorded and categorized.
- Data was analyzed using percentage distribution and visual charts.

PILOT STUDY

Before conducting the main survey, we performed a pilot study with 5 participants.

The purpose was to:

- Test whether the song triggered noticeable emotional reactions
- Observe spontaneous verbal and non-verbal responses
- Identify recurring words or patterns in description
- Refine unclear or repetitive questions

Observations from Pilot Study:

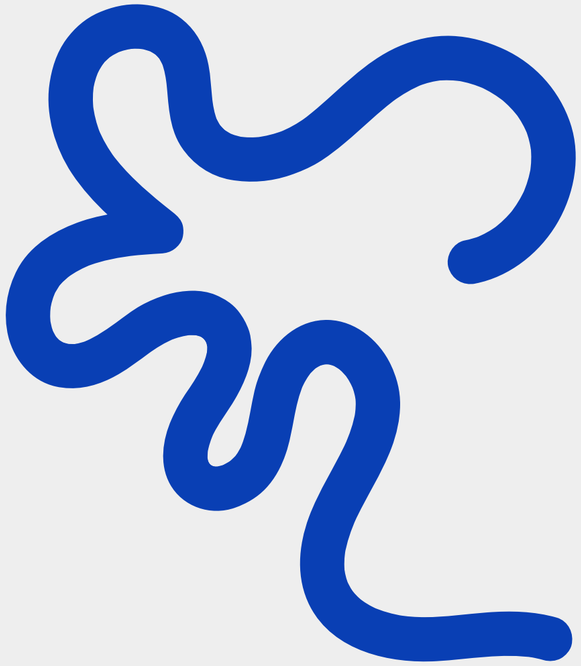
- Participants used words like “intense”, “dramatic”, “cold”, and “cinematic” repeatedly
- Most reported a change in emotion during the build-up
- Many associated the song with movies or high-pressure situations
- Open-ended answers were too varied, making analysis difficult

Outcome:

Based on these insights, we:

- Converted many questions into multiple-choice format
- Added temperature perception (Hot/Cold)
- Included spatial perception (Small/Medium/Large space)
- Structured questions into Before and After listening sections

The pilot study helped refine and strengthen the final questionnaire.



QUESTIONS ASKED

Section A – Before Listening

1. **How are you feeling right now?**
   - Very low
   - A little low
   - Neutral
   - Good
   - Very good

1. **How often do you listen to music?**
   - Rarely
   - Sometimes
   - Often
   - Very often

Section B – After Listening

1.Did the song remind you of anything?
   - A memory
   - A movie/show
   - A place
   - A person
   - Nothing specific

2.What type of situation does the song feel connected to?
   - Calm
   - Dramatic/Intense
   - Motivational
   - Sad
   - Scary/Thrilling

QUESTIONS ASKED

Section B – After Listening

3.How would you describe the pace of the song?

- Slow
- Medium
- Fast
- Very fast

4.Did your emotions change while listening?

- No change
- Slight change
- Changed a lot

5.What kind of workplace setting fits this song?

- Office
- Creative field
- Emergency setting
- Sports/Gym
- None

6.If this song were a brand, what type would it be?

- Luxury
- Tech
- Entertainment
- Wellness
- Sports

7.If this song were food, what would it be like?

- Sweet
- Spicy
- Bitter
- Strong (like coffee)
- Light/Fresh

QUESTIONS ASKED

Section C – Rapid Fire

1.Does the song feel more:

- Hot
- Cold

2.Would you listen to it again voluntarily?

- Yes
- Maybe
- No

3.How memorable was it after one listen?

- Slightly memorable
- Moderately memorable
- Very memorable
- Extremely memorable

4.The music felt like it filled:

- A small space
- A medium space
- A large/open space

5.Overall intensity:

- Slightly intense
- Very intense
- Extremely intense

LISTENING DURATION

Listening Duration Distribution

- 10 participants listened for around 1 minute
- 8 participants listened for around 1 minute 15 seconds
- 5 participants listened for around 45 seconds
- 2 participants listened for around 1 minute 30 seconds

Final Average Listening Time: Approximately 1 minute 4 seconds

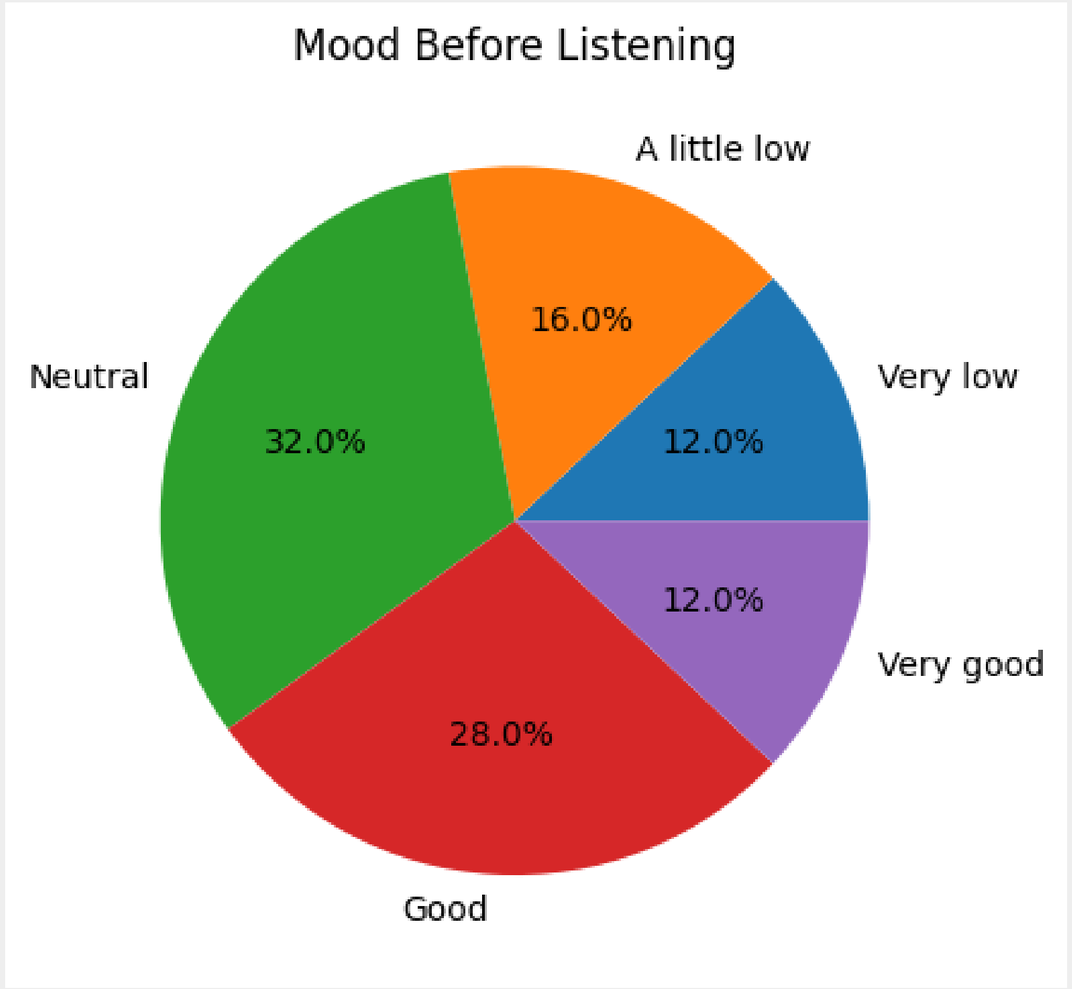
The average listening duration was approximately 1 minute and 4 seconds, with most participants listening between 1 minute and 1 minute 15 seconds before responding to the survey.

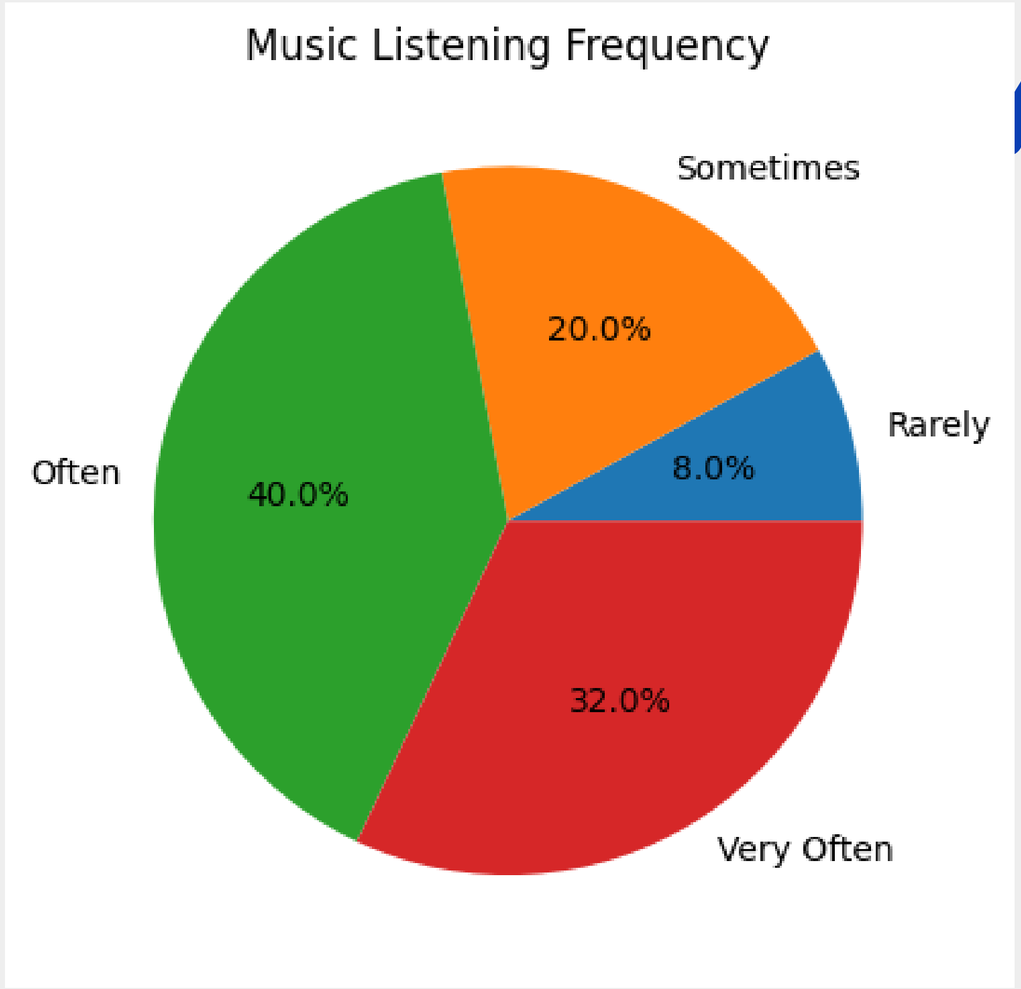
RESULT ACQUIRED

No.	Mood Before	Music Frequency	Reminded Of	Situation Type	Workplace Fit	Brand Type	Food Type	Pace	Emotion Change	Hot/Cold	Listen Again	Memorability	Space Felt	Intensity
1	Neutral	Often	Movie/Show	Dramatic	Emergency	Tech	Strong	Fast	Yes (Calm > Anxious)	Cold	Yes	Very	Large	Extremely
2	Good	Very Often	Movie/Show	Dramatic	Creative	Entertainment	Bitter	Very Fast	Yes	Cold	Yes	Extremely	Large	Extremely
3	A little low	Sometimes	Person	Dramatic	Office	Luxury	Strong	Fast	Slight	Cold	Maybe	Moderate	Medium	Very
4	Neutral	Very Often	Nothing	Motivational	Sports	Sports	Spicy	Very Fast	Yes	Hot	Yes	Very	Large	Extremely
5	Very low	Often	Memory	Sad	Creative	Luxury	Bitter	Medium	Yes	Cold	Yes	Extremely	Large	Very
6	Neutral	Rarely	Memory	Dramatic	Emergency	Luxury	Strong	Fast	Slight	Cold	No	Moderate	Medium	Very
7	Good	Sometimes	Movie	Scary	None	Entertainment	Sweet	Fast	Slight	Cold	Maybe	Very	Small	Very
8	A little low	Very Often	Place	Motivational	Office	Tech	Strong	Fast	Yes	Hot	Yes	Very	Large	Extremely
9	Neutral	Often	Nothing	Dramatic	Office	Wellness	Bitter	Medium	Slight	Cold	Yes	Moderate	Medium	Very
10	Good	Very Often	Movie	Dramatic	Creative	Luxury	Bitter	Fast	Yes	Cold	Yes	Extremely	Large	Extremely
11	A little low	Often	Memory	Dramatic	Office	Luxury	Strong	Fast	Yes	Cold	Yes	Very	Medium	Very
12	Good	Very Often	Movie	Dramatic	Creative	Luxury	Bitter	Medium	Slight	Cold	Yes	Extremely	Large	Very
13	Neutral	Very Often	Movie	Scary	None	Entertainment	Spicy	Very Fast	Yes	Cold	Yes	Very	Large	Extremely
14	Neutral	Often	Place	Motivational	Office	Tech	Strong	Medium	Slight	Cold	Yes	Moderate	Large	Very
15	Very low	Sometimes	Memory	Sad	None	Wellness	Sweet	Medium	Yes	Cold	Maybe	Moderate	Medium	Very
16	Neutral	Often	Nothing	Dramatic	Office	Wellness	Bitter	Medium	Slight	Cold	Yes	Moderate	Medium	Very
17	Good	Very Often	Movie	Dramatic	Creative	Entertainment	Spicy	Very Fast	Yes	Hot	Yes	Very	Large	Extremely
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23	Neutral	Often	Place	Dramatic	Creative	Tech	Strong	Medium	Slight	Cold	Yes	Moderate	Large	Very
24	A little low	Very Often	Place	Dramatic	Office	Tech	Strong	Very Fast	Yes	Cold	Yes	Very	Large	Extremely
25	Neutral	Often	Nothing	Motivational	Wellness	Wellness	Light	Medium	Slight	Cold	Maybe	Moderate	Large	Very

PIE CHART REPRESENTATION

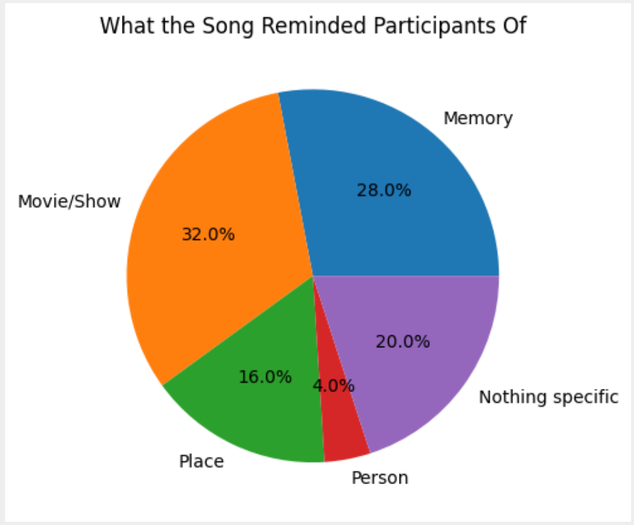
SECTION – A

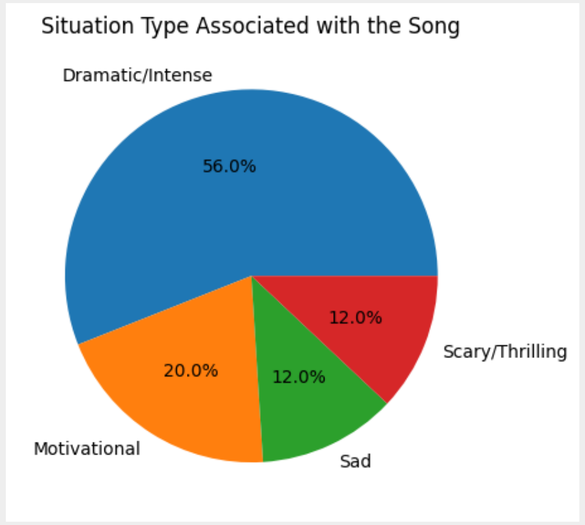


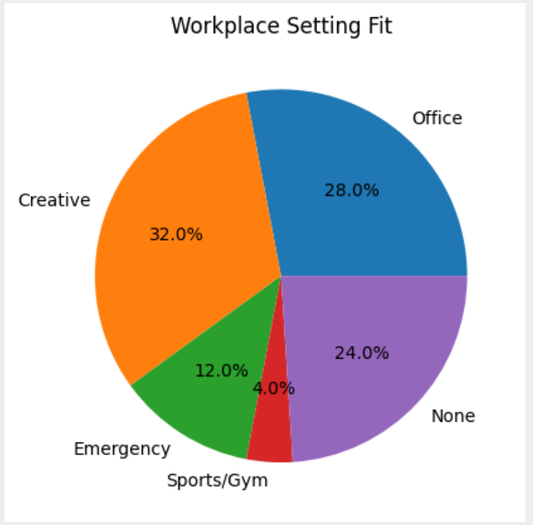


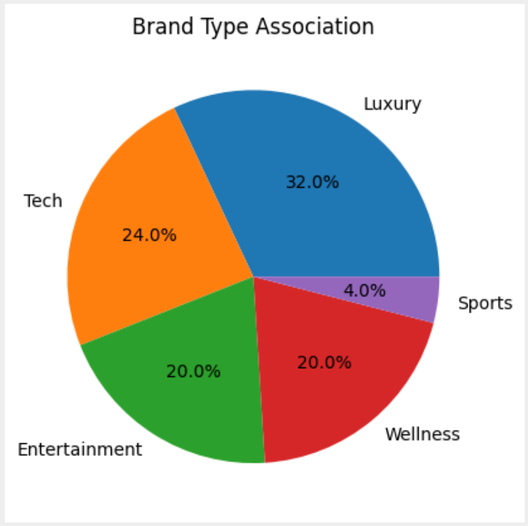
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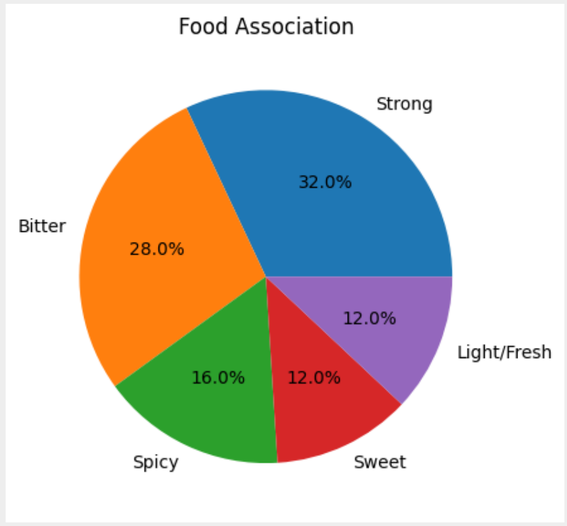
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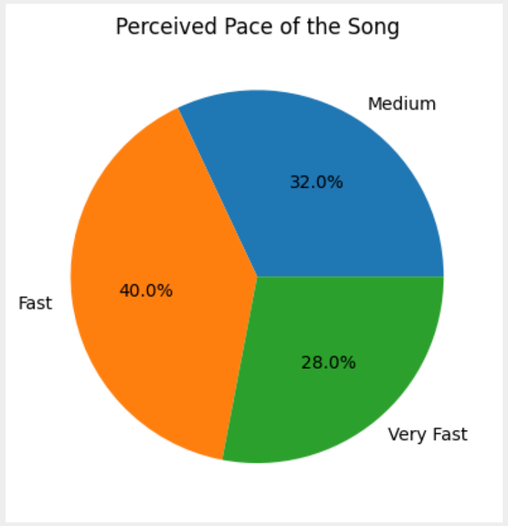


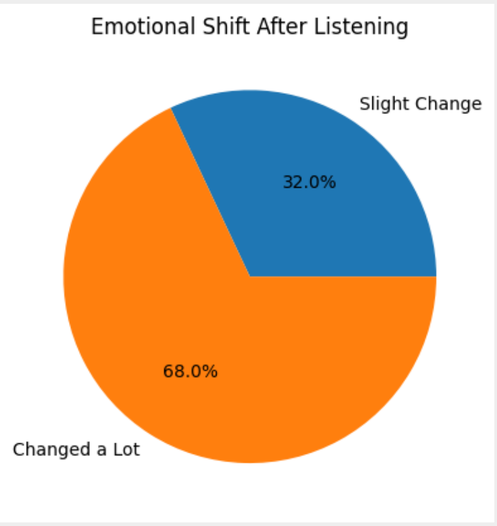






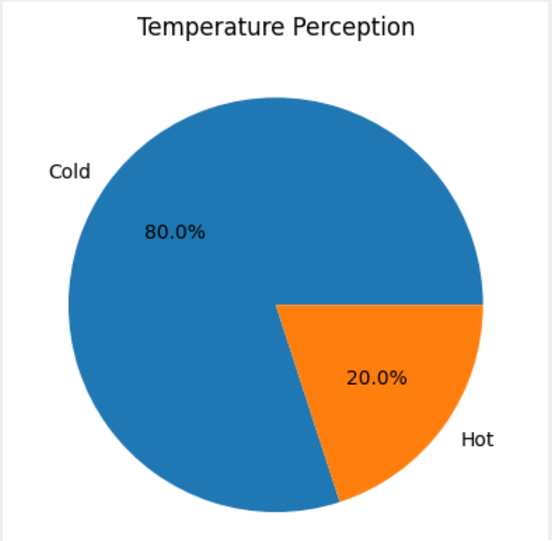


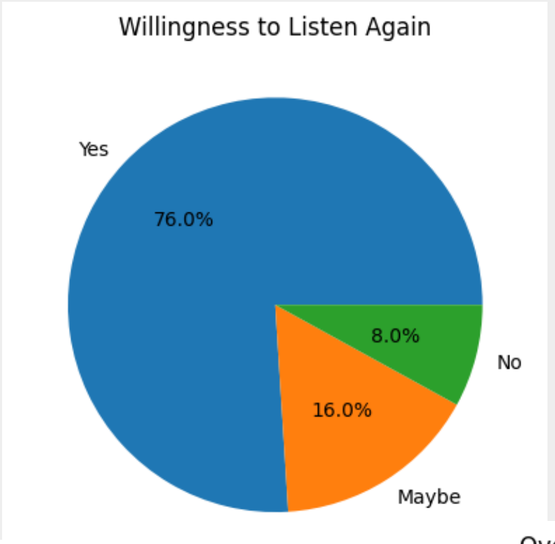


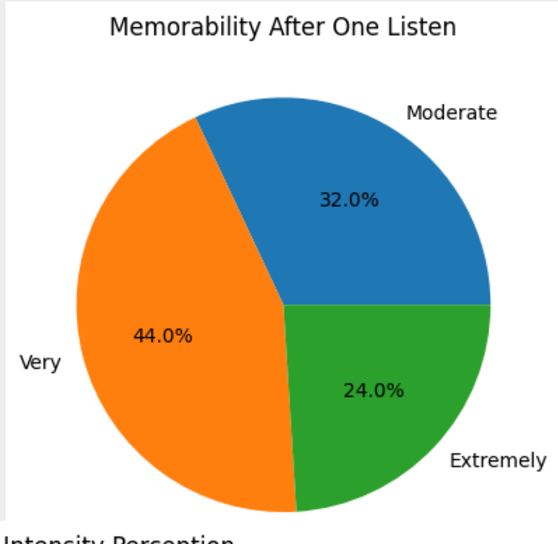


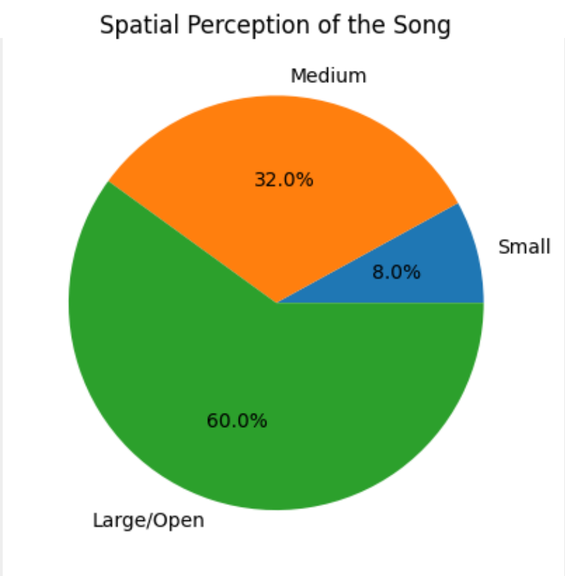
PIE CHART REPRESENTATION

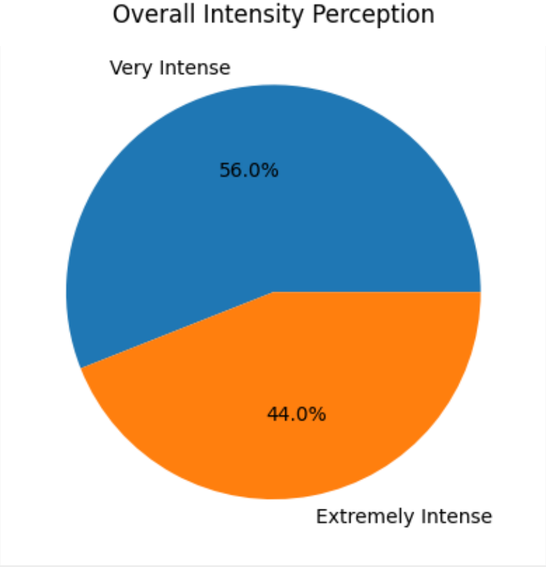
SECTION – C











METHOD OF ANALYSIS

1. Percentage Analysis

Each response category was converted into percentage form to identify dominant patterns.
Example:

- 68% reported strong emotional change.
- 80% perceived the song as “Cold”.
- 60% felt it filled a large/open space.

This helped identify majority trends.

2. Comparative Pattern Analysis

Responses were compared across:

- Emotional vs Spatial perception
- Pace vs Intensity perception
- Brand vs Situation association
- Mood before vs Emotional shift after listening

This helped identify psychological relationships in perception.



GENERAL OBSERVATIONS

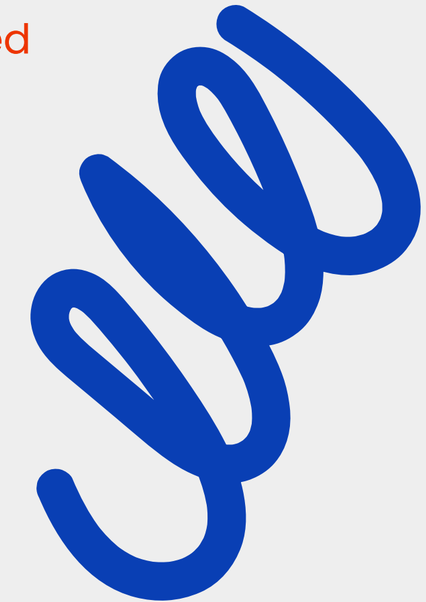
Behavioral Observations

While participants were listening to the song, the following patterns were observed:

- Many participants became visibly still and focused
- Several showed facial tension during the build-up
- Some participants leaned forward or closed their eyes
- A few tapped their fingers subtly with the rhythm
- Participants who initially appeared relaxed showed alert body posture as the intensity increased
- Most avoided distractions during the rising crescendo

Attention & Engagement

- Even those who listened for around one minute appeared fully engaged
- Very few participants showed signs of boredom
- Emotional reactions were more visible during the latter half of the excerpt



CONCLUSION OVERVIEW

Key Findings from the Study

The survey revealed that even brief exposure to the song significantly influenced emotional state, perception, and associative thinking among participants. Despite an average listening time of approximately one minute, strong psychological responses were observed.

Emotional Impact Analysis

Emotional Shift

- 68% reported a strong emotional change
- 32% reported a slight change
- 0% reported no change

Analysis:
The song has a high emotional activation capacity. Its gradual build-up and increasing intensity trigger noticeable emotional shifts, even in short listening durations.

CONCLUSION OVERVIEW

Perception of Intensity & Pace

Intensity
- 100% rated it Very or Extremely intense

Pace
- 68% rated it Fast or Very Fast

Analysis:
The rising tempo and repetitive orchestral layering create urgency and tension. The structure of the composition strongly contributes to perceived intensity.

Temperature & Spatial Perception

Temperature
- 80% described the song as “Cold”

Space
- 60% felt it filled a Large/Open space

Analysis:
Participants associated the song with vastness and emotional distance. The orchestral reverb and dramatic build contribute to a cinematic, expansive feeling rather than a warm or intimate one.

CONCLUSION OVERVIEW

Imagery & Associations

Most Common Associations:

- 32% → Movie/Show
- 56% → Dramatic/Intense situations
- Frequently linked to Luxury and Tech brands
- Food associations leaned toward Strong/Bitter flavors

Analysis:
The song triggered cross-sensory associations. Participants connected sound to visuals (movies), branding identity (luxury/tech), and even taste (strong/bitter), showing how music influences multi-sensory perception.

Behavioral Insight

Even with an average listening duration of approximately 1 minute:

- High memorability
- High replay willingness (76%)
- Strong emotional activation

Interpretation:
The composition's structure allows rapid emotional engagement. It does not require full-length exposure to create psychological impact.

FINAL ANALYSIS

A majority of respondents described the song as dramatic, intense, and emotionally powerful. Most participants reported experiencing a noticeable emotional shift while listening, indicating that the composition successfully altered or heightened their emotional state within a short duration. Even though the average listening time was approximately one minute, the song was still perceived as highly intense and memorable.

The music was predominantly associated with large, open spaces and was commonly described as “cold” in tone. This suggests that participants perceived the song as expansive, cinematic, and emotionally distant rather than warm or comforting. The fast pace and escalating structure of the composition contributed significantly to feelings of urgency and tension.

Participants frequently connected the song to movies, dramatic situations, luxury or tech brands, and strong or bitter flavors such as coffee. This indicates cross-sensory association, where sound influences perception beyond hearing, extending into taste, brand identity, and spatial imagination.
Overall, the findings demonstrate that even a short exposure to a powerful musical piece can influence mood, trigger imagery, and create consistent patterns of perception among different individuals. The study highlights the strong psychological and sensory impact that music can have on human cognition and emotional processing.

THANK YOU